

INPPL1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes INPPL1 gene fusions from the msk_impact_50k_2026 study, covering 14 fusion events. 4/14 fusions (28.6%) were in-frame. 7/14 fusions (50.0%) retained the Endonuclease/Exonuclease/phosphatase family domain. Domain retention was tested with Fisher's exact test ($p=0.048951$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for INPPL1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 14 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was FCHSD2, observed in 4 events.

Domain retention

Endonuclease/Exonuclease/phosphatase family domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 4/4 (100.0%) of in-frame fusions retained the domain, $p=0.048951$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.108911.

Domain disruption

Disruption of the SH2 domain was tested with Fisher's exact test comparing in-frame fusions against all others; 0/4 (0.0%) of in-frame fusions disrupted the domain, $p=1$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

Cutpoint detection

Cutpoint detection scanned 13 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 560 aa (exon 14; permutation-corrected $p=0.033966$, statistically significant at $\alpha=0.05$). This is 131 aa from the nearest configured domain boundary at 429 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=7$) and "not_retained" ($n=6$). For Frame_status, retained: 4/5 (80.0%, 95% CI 37.6%-96.4%); not_retained: 0/2 (0.0%, 95% CI 0.0%-65.8%). Welch's t-test comparing Tumor_variant_count between groups gave means of 19.7143 vs 43.3333, $p=0.434221$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (11 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for INPPL1 in this cohort; expression-association analysis was skipped.

Joint partner

Joint partner reported: INPPL1 has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence reported: INPPL1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (18 row(s) in window_scan, 5 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ANK2	1	0.07142857142857142	0.09629286886244175	0.0	1.0	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2332087369392623	1
SHANK2	1	0.07142857142857142	0.09629286886244175	0.0	0.956	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2266087369392623	2
FCHSD2	4	0.2857142857142857	0.09629286886244175	0.0	0.51	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.22399445122497658	3
PTCH1	1	0.07142857142857142	0.09629286886244175	0.0	0.608	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1744087369392623	4
GAB2	1	0.07142857142857142	0.09629286886244175	0.0	0.31999999999999995	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.13120873693926227	5
NLRP6	1	0.07142857142857142	0.09629286886244175	0.0	0.29200000000000004	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1270087369392623	6
CLPB	1	0.07142857142857142	0.09629286886244175	0.0	0.246	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.12010873693926229	7
FOLR2	1	0.07142857142857142	0.09629286886244175	0.0	0.14600000000000002	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1051087369392623	8
FAT3	1	0.07142857142857142	0.09629286886244175	0.0	0.10799999999999998	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.09940873693926229	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SLC67A1	1	0.07142857142857142	0.09629286886244175	0.0	None	0.37706948295080434	confidence_certainty, domain_disruption, domain_retention, recurrence	0.09789263169324977	10
LRTOMT	1	0.07142857142857142	0.09629286886244175	0.0	0.0	0.37706948295080434	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.0832087369392623	11

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
114	0	7	1	5	0.0	0.46153846153846156	0.3357921019231931
133	0	7	2	4	0.0	0.1923076923076923	0.7160033436347992
183	1	6	2	4	0.3333333333333333	0.5594405594405595	0.2522460504731182
206	1	6	3	3	0.16666666666666666	0.26573426573426573	0.5755524408482516
220	1	6	4	2	0.08333333333333333	0.10256410256410256	0.9890046156985368
364	2	5	4	2	0.2	0.2861305361305362	0.5434357913897181
538	2	5	5	1	0.08	0.10256410256410256	0.9890046156985368
560	2	5	6	0	0.0	0.020979020979020976	1.6782147827453995
805	6	1	6	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

0	3
4	6

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf00017	SH2 domain	13	10	0	3	0.0

domain_retention tables

frame_domain_contingency_table

4	3
0	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Endonuclease/Exonuclease/phosphatase family	13	7	2	4	0.17988505747126438

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	8	14	0.5714285714285714

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ANK2	1
CLPB	1
FAT3	1
FCHSD2	4
FOLR2	1
GAB2	1
LRTOMT	1
NLRP6	1
PTCH1	1
SHANK2	1
SLC67A1	1

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
6	206	200	4	9	1	6	0.16666666666666666	0.26573426573426573	0.5755524408482516	EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0042110-T02-IM7-6, EVT-P-0045578-T01-IM6-11
20	220	200	5	8	1	6	0.08333333333333333	0.10256410256410256	0.9890046156985368	EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0042110-T02-IM7-6, EVT-P-0045578-T01-IM6-11
106	206	100	4	9	1	6	0.16666666666666666	0.26573426573426573	0.5755524408482516	EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0042110-T02-IM7-6, EVT-P-0045578-T01-IM6-11
114	214	100	4	9	1	6	0.16666666666666666	0.26573426573426573	0.5755524408482516	EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0042110-T02-IM7-6, EVT-P-0045578-T01-IM6-11
114	314	200	5	8	1	6	0.08333333333333333	0.10256410256410256	0.9890046156985368	EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0042110-T02-IM7-6, EVT-P-0045578-T01-IM6-11
120	220	100	4	9	1	6	0.16666666666666666	0.26573426573426573	0.5755524408482516	EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0045578-T01-IM6-11
133	233	100	4	9	1	6	0.16666666666666666	0.26573426573426573	0.5755524408482516	EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0045578-T01-IM6-11
133	333	200	4	9	1	6	0.16666666666666666	0.26573426573426573	0.5755524408482516	EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0045578-T01-IM6-11
164	364	200	4	9	2	5	0.8	1.0	-0.0	EVT-P-0005693-T03-IM6-34, EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5
183	383	200	4	9	2	5	0.8	1.0	-0.0	EVT-P-0005693-T03-IM6-34, EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5
605	805	200	4	9	4	3	None	0.06993006993006994	1.1553360374650619	EVT-P-0004226-T03-IM6-9, EVT-P-0004226-T04-IM6-8, EVT-P-0004226-T07-IM7-10, EVT-P-0004226-T08-IM7-7
705	805	100	4	9	4	3	None	0.06993006993006994	1.1553360374650619	EVT-P-0004226-T03-IM6-9, EVT-P-0004226-T04-IM6-8, EVT-P-0004226-T07-IM7-10, EVT-P-0004226-T08-IM7-7

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
755	805	50	4	9	4	3	None	0.06993006993006994	1.1553360374650619	EVT-P-0004226-T03-IM6-9, EVT-P-0004226-T04-IM6-8, EVT-P-0004226-T07-IM7-10, EVT-P-0004226-T08-IM7-7
780	805	25	4	9	4	3	None	0.06993006993006994	1.1553360374650619	EVT-P-0004226-T03-IM6-9, EVT-P-0004226-T04-IM6-8, EVT-P-0004226-T07-IM7-10, EVT-P-0004226-T08-IM7-7
805	830	25	4	9	4	3	None	0.06993006993006994	1.1553360374650619	EVT-P-0004226-T03-IM6-9, EVT-P-0004226-T04-IM6-8, EVT-P-0004226-T07-IM7-10, EVT-P-0004226-T08-IM7-7
805	855	50	4	9	4	3	None	0.06993006993006994	1.1553360374650619	EVT-P-0004226-T03-IM6-9, EVT-P-0004226-T04-IM6-8, EVT-P-0004226-T07-IM7-10, EVT-P-0004226-T08-IM7-7
805	905	100	4	9	4	3	None	0.06993006993006994	1.1553360374650619	EVT-P-0004226-T03-IM6-9, EVT-P-0004226-T04-IM6-8, EVT-P-0004226-T07-IM7-10, EVT-P-0004226-T08-IM7-7
805	1005	200	4	9	4	3	None	0.06993006993006994	1.1553360374650619	EVT-P-0004226-T03-IM6-9, EVT-P-0004226-T04-IM6-8, EVT-P-0004226-T07-IM7-10, EVT-P-0004226-T08-IM7-7

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
780	805	25	4	9	4	3	None	0.06993006993006994	1.1553360374650619	EVT-P-0004226-T03-IM6-9, EVT-P-0004226-T04-IM6-8, EVT-P-0004226-T07-IM7-10, EVT-P-0004226-T08-IM7-7
20	220	200	5	8	1	6	0.08333333333333333	0.10256410256410256	0.9890046156985368	EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0042110-T02-IM7-6, EVT-P-0045578-T01-IM6-11
106	206	100	4	9	1	6	0.16666666666666666	0.26573426573426573	0.5755524408482516	EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0042110-T02-IM7-6, EVT-P-0045578-T01-IM6-11
120	220	100	4	9	1	6	0.16666666666666666	0.26573426573426573	0.5755524408482516	EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5, EVT-P-0045578-T01-IM6-11
164	364	200	4	9	2	5	0.8	1.0	-0.0	EVT-P-0005693-T03-IM6-34, EVT-P-0009791-T02-IM6-12, EVT-P-0024858-T01-IM6-36, EVT-P-0031523-T02-IM6-5

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007343-T02-I M6-4	P-0007343-T02-I M6		Prostate Cancer	PRA D	INPP L1-A NK2	ANK2	unknown	five_prime	71943348	14	560	False	560	True	exact match	disrupted	0.45517241379310347	True	SH2 domain		Endonuclease/Exonuclease/phosphatase family	INPP L1 (NM_001567) - ANK2 (NM_001148) rearrangement: t(4;11)(q25;q26;q13.4)(chr4:g.113898703:chr11:g.71943348) Protein Fusion: mid-exon {INPPL1: ANK2}	NA
EVT-P-0031523-T02-I M6-5	P-0031523-T02-I M6		Breast Cancer	BRC NOS	INPP L1-CL PB	CLPB	unknown	three_prime	71940164	5	183	False	183	True	exact match	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family	SH2 domain		INPP L1 (NM_00156) rearrangement: c.549:INPPL1_chr11:g.72024501 del Antisense Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 2110-T02-I M7-6	P-004 2110-T02-I M7		Bladder Cancer	BLCA	INPP1-FA T3	FAT3	unknown	five_prime	7193 9487	3	114	False	114	True	exact match	lost	0.0	False	SH2 domain	Endonuclease/ Exonuclease/ phosphatase family		INPP1 (NM_001567) - FAT3 (NM_001008781) rearrangement: c.342:INPPL1_c.3669+26772:FAT3del Protein Fusion: mid-exon {INPP1:FA T3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004226-T08-IM7-7	P-0004226-T08-IM7		Bladder Cancer	UTUC	INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945806	21	805	True	805	True	exact match	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family; SH2 domain			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+147:INPPL1_c.705+16442:FCHSD2inv Protein Fusion: in frame {INPPL1:FC HSD2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004226-T04-IM6-8	P-0004226-T04-IM6		Bladder Cancer	UTUC	INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945806	21	805	True	805	True	exact match	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family; SH2 domain			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+147:INPPL1_c.705+16442:FCHSD2inv Protein Fusion: in frame {INPPL1:FC HSD2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004226-T03-IM6-9	P-0004226-T03-IM6		Bladder Cancer	UTUC	INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945742	21	805	True	805	True	exact match	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family; SH2 domain			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+83:INPPL1_c.705+16440:FC HSD2 inv Protein Fusion: in frame {INPPL1:FC HSD2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004226-T07-1M7-10	P-0004226-T07-1M7		Bladder Cancer	UTUC	INPPL1-FCHSD2	FCHSD2	in-frame	five_prime	71945792	21	805	True	805	True	exact match	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family; SH2 domain			INPPL1 (NM_001567) - FCHSD2 (NM_014824) rearrangement: c.2415+133:INPPL1_c.705+16441:FCHSD2inv Protein Fusion: in frame {INPPL1:FC HSD2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045578-T01-IM6-11	P-0045578-T01-IM6		Ovarian Cancer	HGSO	INPPL1-FOLR2	FOLR2	out-of-frame	five_prime	71939641	3	133	True	132	True	hybrid junction condition: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	SH2 domain	Endonuclease/Exonuclease/phosphatase family		INPPL1 (NM_001567) - FOLR2 (NM_001113536) rearrangement: c.397+99:INPPL1_c.151-191:FOLR2dup Protein Fusion: out of frame {INPPL1:FOLR2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009791-T02-IM6-12	P-0009791-T02-IM6		Breast Cancer	IDC	INPP L1-G AB2	GAB2	out-of-frame	five_prime	71940493	5	220	True	219	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	SH2 domain	Endonuclease/Exonuclease/phosphatase family		INPP L1 (NM_001567) - GAB2 (NM_080491) rearrangement: c.660-16:INPP L1_c.75+18325:GAB2 inv Protein Fusion: out of frame {INPP L1:G AB2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 8151-T01-I M6-3 1	P-004 8151-T01-I M6		Endometrial Cancer	UCC C	INPP L1-LR TOM T	LRTO MT	unknown	five_prime	7194 8466	26	1060	False	1059	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Endonuclease/Exonuclease/Phosphatase family; SH2 domain			INPP L1 (NM_001567) - LRT OMT (NM_145309) rearrangement: c.3178:1NPPL1_c.83-1241:LRT OMT dup Protein Fusion: mid-exon {INPPL1: LRTO MT}	NA

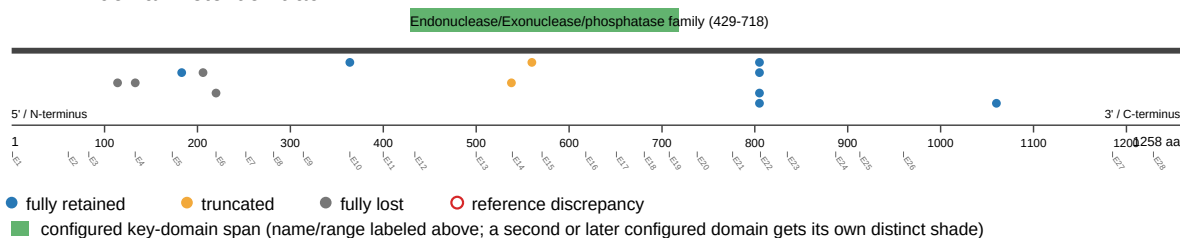
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005693-T03-IM6-34	P-0005693-T03-IM6		Glioma	EPM	INPPL1-PTCH1	PTCH1	out-of-frame	three_prime	71941318	10	364	True	364	True	exact match	retained	1.0	False	Endonuclease/Exonuclease/phosphatase family	SH2 domain		PTCH1 (NM_000264)-INPPL1 (NM_001567) rearrangement: t(9;11)(q22.32;q13.4)(chr9:g.98211830::chr11:g.71941318) Protein Fusion: out of frame {PTCH1:INPPL1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002-4858-T01-IM6-36	P-002-4858-T01-IM6		Non-Small Cell Lung Cancer	LUSC	NLRP6-INPPL1	NLRP6	unknown	five_prime	71940232	5	206	False	205	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	SH2 domain	Endonuclease/Exonuclease/phosphatase family		INPPL1 (NM_001567) - NLRP6 (NM_138329) rearrangement: c. 617:INPPL1_c.2138:NLRP6dup Protein Fusion: mid-exon {INPPL1: NLRP6}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 1882-T01-I M6-37	P-003 1882-T01-I M6		Breast Cancer	BRCNOS	SHANK2-INPPL1	SHANK2	unknown	three_prime	71942656	13	538	False	538	True	exact match	disrupted	0.6241379310344828	True		SH2 domain	Endonuclease/Exonuclease/phosphatase family	INPPL1 (NM_001567) rearrangement: c. 1612:INPPL1 chr11:g. 70501515 dup Antisense Fusion	NA

domain retention outliers

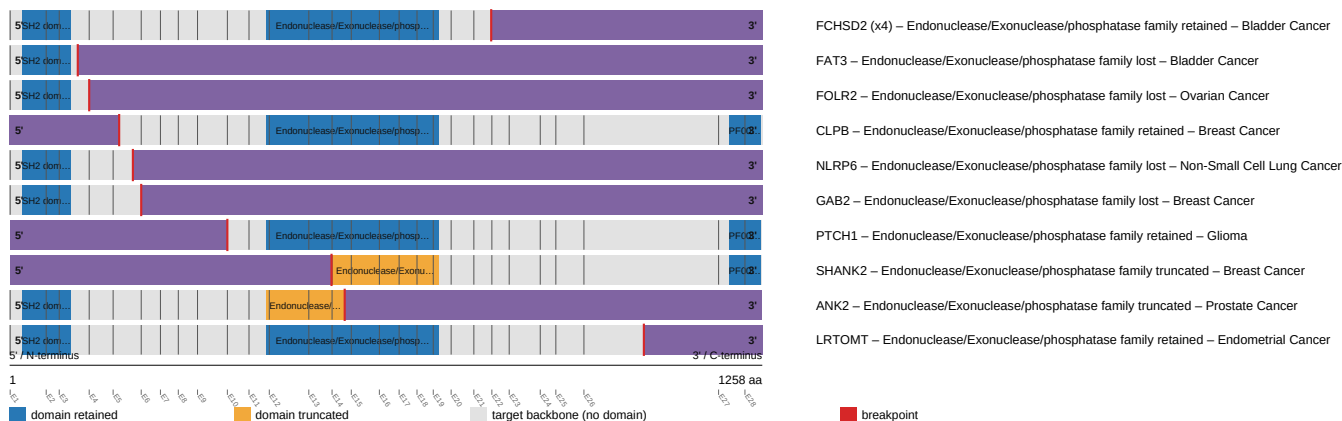
INPPL1 domain-retention track



fusion schematic

INPPL1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (INPPL1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative: row labels identify partners.

reference comparison

Reference vs msk_impact_50k_2026

